DeepDock: Enhancing Ligand-protein Interaction Prediction by a Combination of Ligand and Structure Information

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***Abstract*— The prediction of precise protein-ligand binding activities can accelerate drug discovery by virtual screening--a computational technique that predicts whether a small molecule ligand is able to bind to a specific target. Thus, it is crucial to improve the performance of virtual screening. However, previous models for solving this problem are either ligand-based or structure-based. In this paper, we propose a universal deep neural network model called DeepDock that predicts protein-ligand interaction by using both ligand and structure information. Using the combination of two types of information, our model consists of embedding, convolution, max pooling, and fully-connected layers. In particular, different types of inputs are concatenated before being fed into the fully-connected layers. In the experiments, we compare our approach to the competing methods against two benchmark datasets under different settings. The experiment results have demonstrated that DeepDock can improve predictive performance by more than 4% on both DUD-E and MUV datasets in terms of AUPR**

Keywords—virtual screening, ligand-protein interaction, structure-based, ligand-based, deep learning, neural network.

I. Introduction

Drug discovery aims to find such compounds with specific properties that they can have active interactions with protein targets. As such, they can cure corresponding diseases. However, it is difficult to find new drugs or reposition approved drugs. This is because drug discovery needs plenty of laboratory experiments and numerous clinical trials [1]. Accelerating the process of drug discovery is valuable. In particular, computational approaches, or in silico approaches are thought to be effective as they avoid biomedical experiments by filtering a large number of compounds. This search process is known as virtual screening (VS) [2].

In the past few years, many researchers have made use of virtual screening to discover new compounds with pharmaceutical potentials. In general, virtual screening has two main categories: ligand-based and structure-based [3]. The ligand-based approaches rely on analysing physicochemical properties and descriptors of ligands, including molecular weight, octanol/water partition coefficients, and fingerprints. According to these descriptors, candidate ligands are compared with approved drugs and their activities are then predicted. There are many freely available online chemical databases of molecules, such as ChEMBL [4], DrugBank [5], and PubChem [6]. It is easy to obtain descriptors of candidate ligands. Thus ligand-based approaches have been widely adopted [7]. On the other hand, if the structures of ligand-target complexes are known, the structure-based approaches can be used to determine whether there is an active interaction between a given ligand-target pair or not. Structural information with high resolution is normally measured by physicochemical experiments, such as nuclear magnetic resonance (NMR) and X-ray crystallography diffraction (XRD). As such, there is a limited number of experimentally measured structures of ligand-target complexes. For dealing with large-scale virtual screening, the structural information is in fact simulated computationally by molecular docking programs. The goal of molecular docking is to predict how a small molecule ligand will be bound to a protein target. Molecular docking programs first attempt to find out possible binding modes of ligand-protein complexes by heuristic search algorithms, and then use scoring functions to estimate the binding affinities (energy scores) of these modes. Molecular docking has the advantage of interpretability, but it is highly time-consuming. The binding conformations given by molecular docking programs can approximate the corresponding real co-crystal structures of ligand-target complexes. However, it is still a crucial challenge for computational researchers to perform better on the task of binding energy prediction [8].

A vital step in virtual screening is to extract features for ligand compounds and target receptors. Ligand-based information includes a variety of ligand-molecular descriptors, which are used as their inputs by many early algorithms. This is because it is easy to collect and handle the ligand-based data. These algorithms model the quantitative structure-activity relationships (QSAR) or discern pharmacophores of ligand compounds for virtual screening [9-12]. As simple compound descriptors, physicochemical properties directly affect the pharmacokinetics and toxicities of ligands. In addition, as frequently-used ligand-based descriptors, molecular fingerprints characterize the topological characteristics of molecules. Traditional feature extraction approaches depend on complex expert knowledge, which may lead to the loss of relevant information and are not applicable for universal tasks. Recently, deep learning has achieved success in a variety of fields [13]. In particular, deep learning can learn features directly from the raw data of different tasks [14]. Based on Extended-Connectivity Fingerprints (ECFP) [15], for example, Duvenaud et al. built

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convolutional networks on graphs to learn differentiable molecular fingerprints and showed that these data-driven features had better predictive performance [16]. In fact, the binding between a ligand molecule and a target protein receptor is a process of geometric matching like 'hand-in-glove'. Thus, the three-dimensional structure information of the ligand-target complex plays an important role in virtual screening. The structure-based methods extract features of ligand-target complexes from their structural information such as atom coordinates, element types, and partial charges. Specifically, artificial neural networks have been used for the structure-based virtual screening to predict drug activities [17-23]. In fact, already available web applications enable users to submit ligand-target samples for prediction. This is suitable for large-scale virtual screening [22-23].

In summary, previous works on virtual screening focus only on one aspect, i.e., either ligand-based or structure-based. However, whether a compound is a potential drug or not depends on both its pharmacokinetics and structural characteristics. In this paper, we propose a novel approach to virtual screening, called DeepDock. Our approach integrates both ligand-based and structure-based information seamlessly. In addition, we utilize a weighted binary cross-entropy loss function to address the problem of the class imbalance in virtual screening. The class imbalance refers to the fact that the number of positive samples (active ligands) is much smaller than that of negative samples (decoys) in drug datasets.

We test our model and compare it with previous methods on two popular datasets for virtual screening of DUD-E and MUV. The experiment results show that the combination of both types of data in DeepDock can improve predictive performance by more than 4% on both DUD-E and MUV datasets in terms of AUPR, compared with the deep learning method of DeepVS, which was based on only structural information proposed by Pereira et al. [21].

II. Methods

A. Overview

As mentioned before, our key idea is to use both ligand-based and structure-based information instead of only one type of information as inputs. Fig. 1 shows an overview of our model, which is detailed in the following sections.

B. Ligand information and Structure Information

We used both ligand-based and structure-based information as the inputs of neural networks.

According to Lipinski's rule of five, an empirical rule that determines whether a compound has the physicochemical properties that make itself an oral drug in humans [24], we took into consideration 7 types of physicochemical properties: molecular weight, the number of H-bond donors, the number of H-bond acceptors, the number of rotatable bonds, the number of aromatic rings, the polar surface area (PSA), and octanol/water partition coefficients (logP). These properties are related closely to the absorption, distribution, metabolism, excretion, and toxicity (ADMET) characteristic of a compound. The number of H-bond donors and that of H-bonds acceptors determine the strength of H-bond interaction, and too many H-bond donors or acceptors would lead to large probability of intramolecular H-bond interaction, which make the ligand unlikely bind to other molecules. The number of rotatable bonds is related to the flexibility of the molecule, which is important for the binding process. Compounds with a large number of aromatic rings might cause strong metabolic toxicity. Molecular weight, the polar surface area and logP have influences on the bioavailability of the ligand. It is usually hard to absorb too heavy or too big molecules, and logP relates to the distribution and transportation between cytosol and cell membranes. To some degree, they can reflect how big the probability of binding to biological targets is. We also used Extended-Connectivity Fingerprints, one of the states of the art in molecular fingerprints as a ligand-based descriptor. In this work, the parameter "radius" was set to 2 while the fingerprint length (hash table size) was set to 1024. As such, we obtained a binary fingerprint vector for each compound ligand (ECFP4). Each bit in the fingerprint indicates whether or not there is a particular substructure occurring in a corresponding ligand molecule. Both physicochemical properties and ECFP4 were calculated by RDKit, an open-source cheminformatics software library (https://www.rdkit.org/).

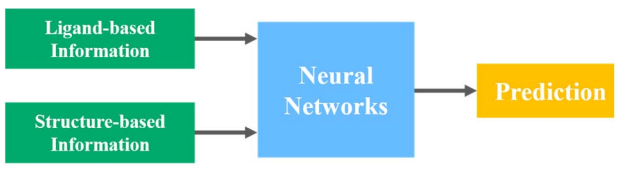

Fig. 1. The overview of our DeepDock model

For using the three-dimensional structural information of the ligand-target complexes, preprocessing need to be conducted. The binding pocket (or binding site), the chemical environment around a ligand molecule when it binds to the target macromolecule, is almost dominant in the binding process and determines the strength of binding. Therefore, we extracted the context of each atom of a ligand molecule as the structure-based input. In the model DeepVS, proposed by Pereira et al., the context of a reference atom "a" is defined by a set of basic attributes extracted from its neighbourhood [21]. In order to study whether the additon of ligand information inpu is effective or not, we adopt their method for structural information extraction to make the structural-based input stay the same in this research. The neighbourhood consists of the number of k_c atoms in a ligand molecule that are closest to "a" (including itself) measured by Euclid distance in space, and the number of k_p atoms closest to "a" in the protein target. Fig. 2 illustrates an example of a reference atom and its neighbourhood. In our work, we set k_c to 6 and k_p to 2. The basic attributes include the atom types, atomic partial charges, amino acid types, and the distances from neighbours to the reference atom. For the basic attributes of atom types and amino acid types, which are discrete string representations, we mapped them to integer indices for convenience. For the basic attributes of atomic partial charges and the distances, it is necessary to discretize them. Therefore, bins were built equidistantly between a pre-defined minimum value and maximum value to perform the discretization. Each basic attribute can be thus represented by an integer index vector with a length of 8 (amino acid types with 2), with each element indicating the corresponding attribute value of an atom in the neighbourhood of the reference atom. For each basic attribute, atoms are arranged by those in a ligand molecule first and then those in the target protein. They are

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finally sorted by their distances to the reference atom, from the closest to the farthest. As a result, the context of an atom in a ligand molecule can be represented by four integer index vectors; that is to say, the chemical environment of the binding pocket of a ligand-target complex can be characterized by four integer index matrices, where each row vector represents one basic attribute information of one atom in a ligand molecule. Fig. 3 shows how to generate inputs for the deep neural network in detail.

C. Model Details

We follow the main architecture of DeepVS model including embedding layers, convolution layers and two fully-connected layers, to ensure that the change of predictive performance is not caused by the difference of neural network frameworks.

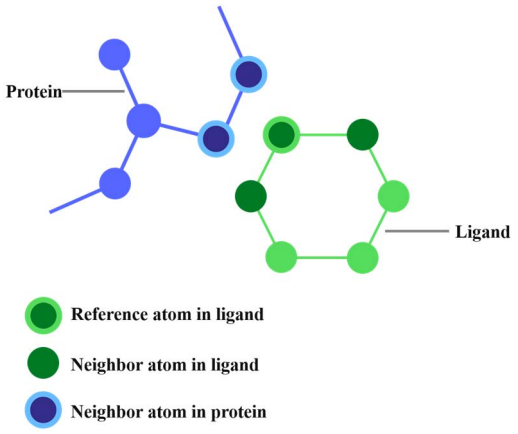

Fig. 2.A reference atom and its neighborhood.

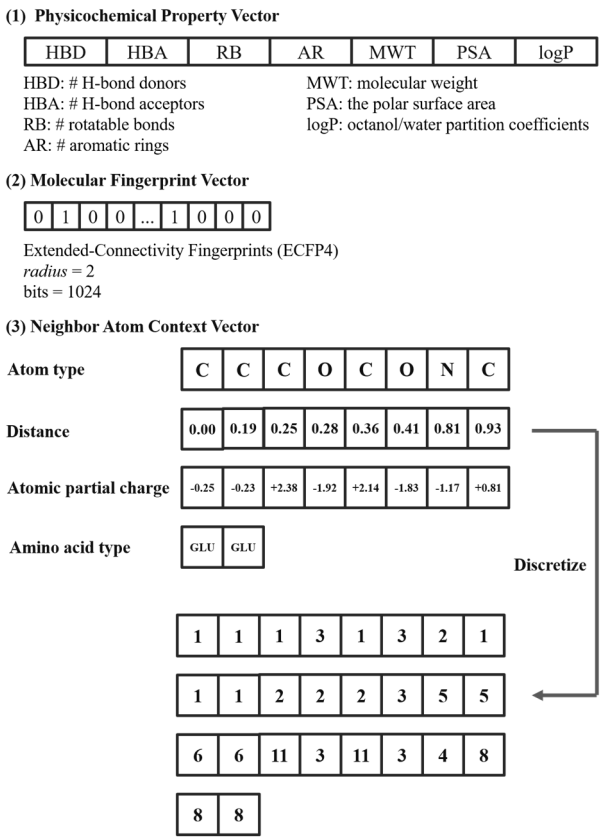

Fig. 3.Input details. (1) and (2) are ligand information, while (3) is generated from the three-dimensional structure of the ligand-target complex.

The first hidden layer in our model consists of four independent embedding layers, aiming to transform respective integer index vectors of basic attributes into real-valued dense vectors of fixed sizes. These embeddings are learned by the model automatically during a training process. Embedding operation can extract more abstract and informative features for the raw input data. It has been proved that this performs well in the field of natural language processing. After embedding layers, four embedding feature vectors are concatenated to form a global feature vector for each atom in a ligand molecule. In this way, the abstract features of a ligand-target complex are represented by a matrix.

A convolution layer with a bias vector and a hyperbolic tangent activation function is then added to our prediction model. The purpose of this is to learn more high-level features from the embedding matrix given by the previous layer. This operation is equivalent to matrix multiplication. Note that different ligand molecules may have a different number of atoms. Therefore, the output matrices of different ligands returned by the second layer may have a different number of rows, but with the same number of columns. For addressing this problem, a max-pooling layer is added immediately after the convolution layer. The Max pooling operation would select the maximum value in the matrix along the vertical axis for each column vector, with removing the other values. As a result, a row vector with a fixed size summarizing the feature information of a ligand-target complex is generated for the following layers.

Next, the structure-based feature vector given by the max-pooling layer is concatenated with the ligand-based feature vector from the input layer. After concatenating, the whole feature vector that contains both structural and ligand information is processed by two fully-connected layers in sequence. The last layer outputs the raw score of the two labels: active ligand and inactive ligand for the input ligand-target complex. To normalize these scores, we utilize the softmax function that transforms them into a probability distribution. Finally, our model returns two probability scores of labels, whose values are between 0 and 1. The higher probability score of an active label indicates that there is more likely an active interaction between an input ligand and its target. Fig. 4 shows the details of our model. The four structral input index matrices $\boldsymbol{X}_i$ (i = atom, distance, charge or aminoacid) are processed by four corresponding embedding layers respectively, transforming into four three-dimensional tensors. We reshape them into matrices $\boldsymbol{E}_i$ with a shape of $N \times d_i$, where N is number of atoms in the ligand and d_i is the product of the embedding length and the size of the atomic neigbourhood. Then these four embedding matrices are concatenated to form the matrix $\boldsymbol{E}$ for convolution. After convolution and max pooling, we get the high-level structural feature vector $\boldsymbol{u}_1$. To make use of ligand information, we concatenate physicochemical property vector $\boldsymbol{input}_{\text{physicochemical}}$ and (or) molecular fingerprint vector $\boldsymbol{input}_{\text{fingerprints}}$ with $\boldsymbol{u}_1$ as follows:

$$\boldsymbol{u}_2 = [\boldsymbol{u}_1 ; \boldsymbol{input}_{\text{physicochemical}}; \boldsymbol{input}_{\text{fingerprints}}], \quad (1)$$

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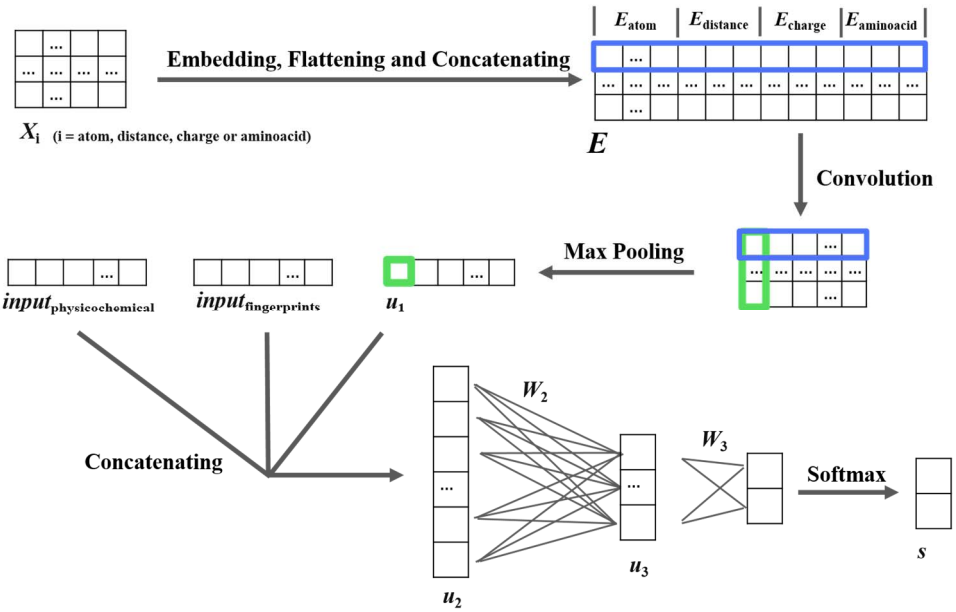

Fig. 4. The details of our DeepDock model. The structural inputs are four basic feature index vectors of each ligand-atom's neighborhood and the ligand inputs are the physicochemical property vector and (or) the molecular fingerprint vector. $\boldsymbol{E}_i$ (i = atom, distance, charge or aminoacid) is the embedding matrix with a shape of $N \times d_i$, where N is number of atoms in the ligand and d_i is the product of the embedding length and the size of the atomic neigbourhood. $\boldsymbol{W}_2$ and $\boldsymbol{b}_2$ are the parameters of the fully connected layer, while $\boldsymbol{W}_3$ and $\boldsymbol{b}_3$ are those of the output layer.

We use a hidden layer and an output layer to produce the final prediction scores as follows:

$$\boldsymbol{s} = \boldsymbol{W}_3(\boldsymbol{W}_2\boldsymbol{u}_2 + \boldsymbol{b}_2) + \boldsymbol{b}_3, \tag{2}$$

$$p_j = \exp(\boldsymbol{s}_j) / \Sigma_j \exp(\boldsymbol{s}_j). \tag{3}$$

where i $\in [0, 1]$ and p_0 and p_1 is the predicted probabilty of a component is negative or positive, respectively.

D. Loss Function

The problem of virtual screening differs from the vast majority of classification problems. This is because the datasets often contain a large number of negative samples but with only a few positive samples. For this class imbalance, we use a weighted binary cross-entropy loss function to reduce the negative effect. Specifically, the weighted binary cross-entropy loss function is written as:

$$L = -[\lambda * y * \log(p_1) + (1 - y) * \log(1 - p_1)] \tag{4}$$

where $y \in \{0, 1\}$ is the true label of a sample, $p_1 \in [0, 1]$ is a probability score of a positive label predicted by the model, and λ is hyperparameter that controls the importance of positive samples. When $\lambda = 1$, this weighted binary cross-entropy loss function degenerates into an ordinary binary cross-entropy loss function. As the value of λ increases, the loss value also increases. So does when the model assigns a high probability score of an inactive label to a true positive sample. The extent of this loss increase is larger than that resulting from a high probability score of an active label being assigned to a true negative sample. Our experiments show that the use of the weighted binary cross-entropy loss function with an appropriate λ value is helpful to improve the prediction performance of our model.

III. Experiments and Results

A. Datasets

Two different datasets were used to evaluate the performance of our virtual screening model, the Directory of Useful Decoys: Enhanced (DUD-E) [25] and the Maximum Unbiased Validation Data Sets (MUV) [26].

As an enhanced and rebuilt version of DUD, the DUD-E dataset was built by Mysinger et al. It provides plenty of challenging decoys for validating virtual screening models. DUD-E contains about 23,000 active compounds and their affinities against 102 targets, with an average of 224 ligands per target. For each active ligand, there are 50 decoys with similar physicochemical properties, such as molecular weight, the number of H-bond donors, and acceptors.

The MUV dataset was designed by Sebastian et al., to make the distances between different active compounds larger than that between active compounds and correspond decoys in chemical space. This dataset prevented both analog bias and artificial enrichment on the point of building datasets for virtual screening. We adopted MUV as the second dataset to further evaluate the performance of our model.

B. Molecular Docking Program

Significantly improving the algorithms of scoring and minimization, Smina is a fork of Autodock Vina developed by Koes et al [8, 27]. Using the pdbqt files, its compound ligand and protein target structure files include partial charge information. Relying on Smina as the molecular docking program, we obtained the three-dimensional structure of ligand-target complexes. For this purpose, we set the parameter of "autobox_add" to 8, the parameter of "exhaustiveness" to 16 and other parameters to default. Note that the docked structure of a ligand-target complex produced by Smina contains more than one binding mode. But we considered only the first one in our experiments, which

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corresponded to the lowest (absolute value was highest) binding energy. In fact, it was the best binding mode.

C. Baseline and Metrics

We compared our DeepDock model to the output energy scores generated by the docking program Smina [8] and the deep learning method of DeepVS, which was based on only structural information proposed by Pereira et al. [21]. A multilayer perceptron (MLP) model that used only ligand-based information as its input was also compared as our baseline.

We randomly split the DUD-E dataset into five parts to perform 5-fold cross-validation on each model. Besides, we also performed leave-one-out cross-validation on the MUV dataset to evaluate the performance. For the reason that the number of targets in MUV dataset is not enough to support 5-fold cross-validation, we needed ample data to guarantee the quality of training. One of the most popular metrics in virtual screening was used to evaluate the performance of each model: the area under the receiver operating characteristic curve (AUROC). The value of AUROC is between zero and one, with a higher value indicating better classification performance. When the value of AUROC is equal to 0.5, it implies that the performance of the corresponding classifier is equivalent to randomly guessing. We also used another metric, the area under the precision-recall curve (AUPR), to evaluate different models. AUPR is more useful in cases where there is an imbalance in the observations between two classes. A higher AUPR means better performance. In addition, enrichment factor (EF) is a commonly used metric for assessing virtual screening as well. $EF_{x\%}$ is defined as follows:

$$EF_{x\%} = (a_{top} / N_{x\%}) / (a / N) \quad (5)$$

where a_{top} is the number of active ligands among the top x% ranking compounds, $N_{x\%}$ is x% of the number of total compounds, a is the number of total active ligands, and N is the number of total compounds. A higher enrichment factor also means better predictive performance. In our experiments, we reported $EF_{10\%}$ for evaluating the model performance.

D. Hyperparameters

For a fair comparison, we used the same hyperparameters of the deep neural network for the baseline method DeepVS. According to the previous work, the hyperparameter values used in our experiments are listed in Table . In order to examine how much the weighted binary cross-entropy loss function could improve the virtual screening performance, we set λ to different values from a set of {1, 20, 50}.

TABLE I. HYPER PARAMETER VALUES OF THE DEEP NEURAL NETWORKS

Hyper Parameter	Annotation	Value
d_{atom}	Atom type embedding length	200
d_{charge}	Charge embedding length	200
$d_{distance}$	Distance embedding length	200
d_{amino}	Amino-acid embedding length	200
h_1	# first dense layer units	400
h_2	# second dense layer units	50

E. Experiments

ADADELTA optimizer was utilized to update the parameters of the networks so that the loss function was minimized [28]. During the training process, we used mini-batches with a size of 128 in each iteration, together with the learning rate of 0.075.

F. Results

1) Predictive Performance: Table shows the comparisons of different methods on predictive performance aginst the DUD-E dataset. The ordinary binary cross-entropy loss function was used in all compared methods. The results indicate that all models surpass the docking program Smina. The average AUROC score of our model with a combination of the ligand-based and structure-based model is slightly higher than that of DeepVS using only structural information. However, the AUPR of the former is 4% higher than that of the latter. More importantly, our model is superior to the only structure-based model by over half targets. To validate this outperformance further, we conducted Student's *t*-test on the average performance of the two methods. The statistical significance level is 0.05 (*p*-value equals 0.034) for AUROC and 0.01 (*p*-value equals 0.005) for AUPR. This indicates that our model can improve predictive performance at a statistically significant level. As a result of the imbalance of the DUD-E dataset, i.e., the number of negative samples (decoys) is much larger than that of positive samples (active ligands), AUROC is not sensitive enough and might be over-optimistic. This is because its computation focuses on both true positive rate and false-negative rate. Any models that produce lower scores to the large majority of negative

TABLE II. PREDICTIVE PERFORMANCE ON DUD-E DATASET (5-FOLD CROSS VALIDATION)

Method	Aver. AUPR	Aver. AUROC	Aver. $EF_{10\%}$
smina	0.108	0.737	3.958
Only structure (DeepVS)	0.446	0.920	7.721
Ligand-based MLP	0.127	0.784	2.867
Structure and ligand physicochemical (DeepDock)	**0.464**	**0.925**	**7.913**

TABLE III. PREDICTIVE PERFORMANCE OF VARIATIONS OF THE NEURAL NETWORKS

Use Physicochemical Properties	Use Molecular Fingerprints	Aver. AUPR	Aver. AUROC	Aver. $EF_{10\%}$
True	True	0.431	0.920	7.773
True	False	**0.464**	**0.925**	**7.913**
False	True	0.401	0.912	7.492
False	False	0.446	0.920	7.721

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samples would reach presentable AUROCs. For the case of the imbalance, AUPR should be regarded as a more important metric, as the computations of precision and recall focus only on how positive samples are predicted. Therefore, our method increases the AUPR. In this way, our method is able to assign more likely correct labels to the interesting positive samples. In addition, our model also achieved the highest $EF_{10\%}$ among the compared methods. This means that our model could enrich the active ligands at the top ranking of total compounds.

On the other hand, ligand-based MLP did not perform well enough, which indicated the importance of structural information. To study whether the addition of ligand-based information was useful, and to exclude the biased information, i.e., the generation of decoys in DUD-E that was based on the molecular fingerprints, we also carried out experiments on models by adding different ligand-based information. Table reports the results. It shows that only the addition of molecular fingerprints does not contribute to the improvement of prediction performance. In contrast, the addition of physicochemical properties helps the model predict better. This can be explained by the following fact. Although the DUD-E is established according to the molecular fingerprints, the molecular fingerprint of a decoy of a target may be similar to that of an active ligand of another target. Besides, a molecular fingerprints is calculated according to the two-dimensional topological characteristic of a molecule, but it cannot reflect three-dimensional structural characteristics when the molecule binds to a target in reality. Two atoms far away from each other in two-dimensional topology may be close for molecule folding casued by hydrophobic or H-bond interactions. However, the physicochemical properties of a ligand determine whether the ligand has good pharmacokinetic quality. So the addition of physicochemical property information could help improve the performance of virtual screening. The results in Table and Table exclude the information leak caused by the building of datasets. The reason for this was that ligand-based MLP and the model with only molecular fingerprints added did not behave better than the model combining structural information and physicochemical property information.

2) Leave-one-out Cross Validation: Table shows the results: the Smina energy scores of the MUV dataset, and leave-one-out cross-validation scores on the MUV dataset of all compared methods. Because the MUV dataset was meticulously designed for challenging virtual screening, the predictive performance of all models degraded significantly. However, the performance of our model was superior to other models. In addition, the two deep learning methods using structural information performed better than the rest ones. Again, this highlights that a combination of structural and ligand information plays an important role in virtual screening. Also, our model is more adaptive when migrating to different datasets.

3) Weighted Binary Cross-Entropy Loss Function: The 5-fold cross-validation performance of our model on the DUD-E dataset is reported in Table , together with a variety of hyperparameter value settings of λ. As shown in the table, the AUROC is insensitive to the variation of λ. When the value λ is 20, the AUPR reaches the highest. Again, for the case in which the number of negative examples is very large in the datasets, AUPR can measure the capacity of a model in predicting positive samples correctly better. The $EF_{10\%}$ is also influenced by the value of λ, achieving the maximum when λ equals 20 as well. This result implies that the weighted binary cross-entropy loss function with an appropriate value of λ could keep a better balance of the trade-off between precision and recall.

TABLE IV. PREDICTIVE PERFORMANCE ON MUV DATASET (LEAVE-ONE-OUT CROSS VALIDATION)

Method	**Aver. AUPR**	**Aver. AUROC**	**Aver. $EF_{10\%}$**
smina	0.002	0.530	1.487
Only structure (DeepVS)	0.012	0.634	3.169
Ligand-based MLP	0.002	0.569	2.867
Structure and ligand physicochemical (DeepDock)	**0.046**	**0.667**	**3.471**

TABLE V. PREDICTIVE PERFORMANCE WITH DIFFERENT λ (DUD-E DATASET)

λ	**Aver. AUPR**	**Aver. AUROC**	**Aver. $EF_{10\%}$**
1	0.464	0.925	7.913
20	**0.470**	**0.926**	**7.956**
50	0.434	0.923	7.769

4) Case Study: For a case study, we took human protein kinase C beta II (DUD-E ID: kpcb, PDB id: 2I0E) as an example. Table reports the predictive performance of different methods on this target. Both of deep learning methods that used structural information performed better than the docking program Smina. However, it is worth mentioning that the ligand-based MLP also achieved passable performance, performing better than Smina. More importantly, our model with the addition of physicochemical property information as its input reached the highest value in every evaluation metric. The deep learning model only using structure ranked a decoy compound (ZINC ID: 40828016) as the 8th (total almost 9000 compounds) while our model ranked it as the 144th. This decoy has 7 H-bond donors and its

TABLE VI. PREDICTIVE PERFORMANCE ON KPCB

Method	**Aver. AUPR**	**Aver. AUROC**	**Aver. $EF_{10\%}$**
smina	0.226	0.781	5.582
Only structure (DeepVS)	0.593	0.958	8.731
Ligand-based MLP	0.261	0.931	7.836
Structure and ligand physicochemical (DeepDock)	**0.730**	**0.969**	**9.104**

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polar surface area is 165, which do not obey the Lipinski's rule and make it unlikely an orally active drug. On the other hand, the model without ligand information ranked an active ligand (ChEMBL ID: 421947) as the 151st , while our model ranked it as the 24th. And all physicochemical properties of this active molecule meet the rule of five. These results further verify the importance of our integration of structural and ligand information.

IV. Conclusion

In this work, we study the extent to which a compound ligand that has the potential to be a drug depends on both physicochemical properties and three-dimensional structure of the ligand-target complex. By integrating ligand information with structural information, we have presented a deep learning network as a novel method for virtual screening. The experiment results have shown that our model could improve the performance of drug virtual screening in terms of AUPR. These results indicate that for virtual screening, it is valuable to take into account the integrated information. Our future work will examine more such information.

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